Wound phantoms: which part is which, and what it measures

Revision C. All dimensions in millimetres. Depth, area and volume are the values engraved on each part, rounded the same way. Revision B prints have the same file names, block sizes and cavities, so they read from this table too; revision B P9 has no fiducial dots (page 2).

Part	STEP file	How to recognise it	Wound shape (mm)	Opening (mm)	Max depth (mm)	Opening area (mm²)	Perimeter (mm)	Volume (mm³)
P1	P1_hemisphere_r15.step	19 mm tall; small round bowl, Ø30	Hemisphere, R 15.000	Ø30.000	15.000	706.86	94.25	7068.58
P2	P2_hemisphere_r25.step	29 mm tall, the tallest block; large round bowl, Ø50	Hemisphere, R 25.000	Ø50.000	25.000	1963.50	157.08	32724.92
P3	P3_cone_r20_d10.step	14 mm tall; round opening Ø40, straight sides to a point	Cone, R 20.000, h 10.000	Ø40.000	10.000	1256.64	125.66	4188.79
P4	P4_paraboloid_30x20_d12.step	16 mm tall; large oval 60 × 40	Elliptic paraboloid, a 30.000, b 20.000, h 12.000	60.000 × 40.000 oval	12.000	1884.96	158.65	11309.73
P5	P5_paraboloid_12x8_d4.step	9 mm tall; small, shallow oval 24 × 16	Elliptic paraboloid, a 12.000, b 8.000, h 4.000	24.000 × 16.000 oval	4.000	301.59	63.46	603.19
P6	P6_flat_plate.step	9 mm tall; flat top, no cavity	None: flat reference	—	0.000	0.00	—	0.00
P7	P7_lobed_r18_d9.step	13 mm tall; flower-shaped opening, uneven lobes	Lobed paraboloid, R0 18.000, h 9.000; 3-lobe amplitude 0.18 R0, 5-lobe amplitude 0.10 R0 at phase 0.7 rad	Lobed, R 13.132–22.868	9.000	1039.45	126.82	4677.55
P8	P8_trough_L40_r9_d8.step	12 mm tall; long slot 58 × 18, rounded ends	Trough: parabolic channel 40.000 long, R 9.000, h 8.000, half-paraboloid ends	58.000 × 18.000 slot	8.000	974.47	136.55	4857.88
P9	P9_paraboloid_on_cyl_r60.step	22 mm tall; top face curved; oval 50 × 36	Elliptic paraboloid, a 25.000, b 18.000, h 10.000, on a cylinder R 60.000	50.000 × 36.000 oval (plan)	10.000	1413.72 skin 1446.60	135.98	7068.58
P10	P10_undermining_base.step	19 mm tall; stepped round pocket Ø56.4 that the lids drop into	Undermined well: chamber Ø40.000 × 12.000 under a Ø24.000 lid opening	Ø24.000 (the lid's hole)	15.000 with a lid	452.39	75.40	16436.81
L1	L1_lid_concentric_r12.step	Disc Ø56 × 3, hole centred	Lid for P10: hole Ø24.000, centred	Sets P10's opening and undermining; values on page 2				
L2	L2_lid_offset6_r12.step	Disc Ø56 × 3, hole 6 mm off-centre	Lid for P10: hole Ø24.000, offset 6.000	Sets P10's opening and undermining; values on page 2				

Blocks are 80 × 80 mm; the height is in “How to recognise it”. **Max depth** is from the top surface to the lowest point of the wound (on P9, from the crown of its curved top). **Opening area** and **perimeter** are measured in plan, looking straight down; on P9, whose top is curved, the area measured on the curved surface itself is given as “skin”. **Perimeter** is not engraved on the parts. P10’s depth, area, perimeter and volume are with either lid seated (page 2).

Measure the printed parts with calipers on arrival and use the measured values as truth: every print process shrinks a little, and the measured number is the one that matters.

P10 undermining set

P10 is a Ø40.000 × 12.000 chamber under a Ø56.400 × 3.000 pocket. A lid drops into the pocket, flush with the top, and its Ø24.000 hole becomes the visible wound; the chamber beyond the hole's edge is undermined. The two lids give the same depth, area and volume and differ only in where the undermining runs.

	P10 with L1 seated	P10 with L2 seated
Lid file	L1_lid_concentric_r12.step	L2_lid_offset6_r12.step
Lid hole	Ø24.000, centred	Ø24.000, centre 6.000 toward 3 o'clock
Max depth (mm)	15.000 (top skin to chamber floor)	15.000 (top skin to chamber floor)
Opening area (mm²)	452.39	452.39
Opening perimeter (mm)	75.40	75.40
Total volume (mm³)	16436.81	16436.81
Visible from above (mm³)	6785.84	6785.84
Hidden under the lid (mm³)	9650.97	9650.97
Undermined area (mm²)	804.25	804.25
Undermining extent (mm)	8.000 all round	2.000 at 3 o'clock to 14.000 at 9 o'clock

Without a lid, the chamber alone, measured from the pocket floor 3.000 mm below the top: opening Ø40.000, area 1256.64 mm², depth 12.000 mm, volume 15079.64 mm³.

Undermining extent at each clock position

From the edge of the lid's hole outward to the chamber wall, along a line from the hole's centre, as a probe would measure it.

Clock	12	1	2	3	4	5	6	7	8	9	10	11
L1 (mm)	8.000	8.000	8.000	8.000	8.000	8.000	8.000	8.000	8.000	8.000	8.000	8.000
L2 (mm)	7.079	4.313	2.578	2.000	2.578	4.313	7.079	10.313	12.970	14.000	12.970	10.313

Orientation. 12 o'clock is the edge of P10 nearest the "P10" engraved on its top; 6 o'clock is the side with the 50.000 key line. On revision C parts, seat the lid with the notch in its rim at the index groove on P10's top, which is at 3 o'clock. Revision B parts have no groove, no notch and no legible "P10": take 6 o'clock as the side with the key line, and seat L2 with its hole nearest 3 o'clock. L1 reads the same at every position.

Scale references on every block's top

Key line: 50.000 mm between the centres of its two end ticks. **Fiducial dots:** four Ø2.500 dots at the corners of a 64.000 mm square, centre to centre. Revision B P9 has no dots: its powder vents cut them away and left four open slots in its top near the corners, so use it for depth, area and volume only, not as a scale reference.

Where the numbers come from

Hemisphere: $V = \frac{2}{3}\pi R^3$, $A = \pi R^2$. Cone: $V = \frac{1}{3}\pi R^2 h$, $A = \pi R^2$. Elliptic paraboloid: $V = \frac{1}{2}\pi abh$, $A = \pi ab$. Lobed paraboloid, $r(\theta) = R0(1 + \Sigma\, a\cos(m\theta + \varphi))$, one term per set of m lobes with amplitude a and phase φ: $V = \frac{1}{4}\pi hR0^2(2 + \Sigma a^2)$, $A = \frac{1}{2}\pi R0^2(2 + \Sigma a^2)$. Trough, channel length L: $V = (4/3)hRL + \frac{1}{2}\pi hR^2$, $A = 2RL + \pi R^2$. P10 with a lid of hole radius r and thickness t: $V = \pi r^2 t + \pi R^2 h$, depth = t + h. Perimeters are integrated numerically from the same outlines.